

Homework 1: Data Science Foundations, Sequencing Formats, QC, and HIV/TB Data Extraction

100 Points Total

Due May 26, 2026

This assignment covers the foundational material from the first few weeks of class: the data science process, sequencing technology and file formats, quality control, and the very first step of the HIV/TB RNA-seq pipeline – downloading the raw reads from SRA and inspecting them with FastQC/MultiQC. All questions are multiple choice and will be administered as a Canvas quiz.

Please save your code/notes for each part of the assignment. Your instructor may request it.

Submit your completed assignment via the designated homework (Quiz) submission site on Canvas.

Course Foundations

1. Which of the following best describes *structured* data?
 - (a) Data organized into rows and columns with a predefined schema (e.g., a gene-by-sample counts matrix or a relational database table)
 - (b) Free-text clinical notes
 - (c) Raw FASTQ sequencing reads
 - (d) Microscopy images

Sequencing Technology

3. Illumina short-read sequencing is based on which chemistry?
 - (a) Sequencing by synthesis (reversible terminator nucleotides imaged after each cycle)
 - (b) Sanger chain-termination with capillary electrophoresis
 - (c) Nanopore translocation of single DNA strands through a protein pore
 - (d) Pyrosequencing on bead-immobilized templates
4. Compared with single-end sequencing, paired-end sequencing has which of the following advantages? (Select the best answer.)
 - (a) Reads come from both ends of each fragment, giving better mapping in repetitive regions and an estimate of insert size
 - (b) It produces longer individual reads (kilobases instead of hundreds of bases)
 - (c) It eliminates the need for adapter trimming
 - (d) It requires no library preparation

5. A study wants to comprehensively call protein-coding variants in 200 individuals on a tight budget. Which sequencing design is the most cost-effective fit?
 - (a) Whole-exome sequencing
 - (b) Deep (30x) whole-genome sequencing
 - (c) Low-coverage whole-genome sequencing
 - (d) Long-read Nanopore sequencing of the entire genome

File Formats and Quality Scores

6. What is the key difference between a FASTA file and a FASTQ file?
 - (a) FASTQ stores a per-base quality score for every base; FASTA does not
 - (b) FASTA is binary; FASTQ is text
 - (c) FASTA contains aligned reads; FASTQ contains unaligned reads
 - (d) FASTQ is used only for proteins; FASTA is used only for nucleotides
7. A base has a Sanger PHRED quality score of $Q = 30$. What is the corresponding probability that the base call is *incorrect*?
 - (a) 1 in 1,000 (0.001)
 - (b) 1 in 100 (0.01)
 - (c) 1 in 10 (0.1)
 - (d) 1 in 10,000 (0.0001)
8. In a standard FASTQ record, the four lines correspond (in order) to:
 - (a) @-prefixed read identifier; the nucleotide sequence; a '+' separator (optionally with the identifier); the per-base quality string
 - (b) Sequence; quality string; identifier; separator
 - (c) Identifier; quality string; separator; sequence
 - (d) Quality string; sequence; identifier; pair flag

HIV/TB Data Extraction and QC

9. You will download the 33 FASTQ files for the HIV/TB dataset from SRA using `sratoolkit`. The SRR accessions are listed in the `SRR_Acc_List_hivtb.txt` file. Which of the following is the correct usage of `fastq-dump` to download a single accession and gzip-compress the output in one step?
 - (a) `fastq-dump -gzip SRR12345678`
 - (b) `fastq-dump -compress gzip SRR12345678`
 - (c) `fastq-dump SRR12345678 -gzip-compress`
 - (d) `fastq-dump -SRR SRR12345678 -gzip`
 - (e) `fastq-dump SRR12345678 | gzip > SRR12345678.fastq.gz`
10. You want to download all 33 HIV/TB FASTQ files *in parallel* on Amarel using a SLURM job array, then run FastQC on every file and aggregate the results with MultiQC. Which of the following statements about that pipeline is *TRUE*?
 - (a) `#SBATCH -array=0-32` creates 33 array tasks indexed 0 through 32, and `multiqc .` aggregates the per-sample FastQC reports into a single interactive summary.

- (b) `#SBATCH -array=0-32` creates 32 array tasks, and MultiQC re-runs quality control with stricter thresholds than FastQC.
 - (c) `#SBATCH -array=33` creates 33 array tasks, and MultiQC trims adapters from the FASTQ files.
 - (d) `#SBATCH -ntasks=33` is the correct directive for parallel array downloads, and MultiQC aligns reads to a reference genome.
11. Go ahead and download all of the HIV/TB FASTQ files into your home directory or scratch directory on Amarel. How large is the SRR6305400 file?